PART II: PATTERNS

Topic 19: Data Quality and Common Pitfalls

The Core Concept

Time series analysis is only as good as the data that feeds it. Public safety data has **unique quality challenges** that can create false signals — apparent trends or spikes that are artifacts of how data was collected, not reflections of what actually happened.

Why It Matters

A data quality problem that goes undetected can lead an entire organization to react to a crisis that does not exist — or to miss one that does. Recognizing these pitfalls is a fundamental analytical skill.

Common Pitfalls in Public Safety Data

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Pitfall	What It Looks Like	What To Do
System / Software Change	A sudden jump or drop with no real-world event	Check if a new RMS was deployed; compare to paper records
Definition Change	Category counts shift because the crime code was reclassified	Review policy memos; note the date of reclassification
Reporting Lag	Recent months look unusually low (records still being entered)	Flag last 1–2 months as provisional; revisit after 30 days
Under-Reporting	A low-crime period that coincides with low staffing	Cross-reference with arrest data or call-for-service logs
Outlier Contamination	One extraordinary event inflates the annual average	Identify, label, and report the outlier separately
Mixing Frequencies	Comparing monthly totals to a daily spike	Always match the time unit before comparing

A Diagnostic Checklist

When you see an unusual spike or drop, ask these questions before reacting:

7. Did we change record-management systems recently?
8. Did the definition or classification of this incident type change?
9. Are recent months' numbers still being entered (reporting lag)?
10. Is there a staffing gap that could explain lower reporting?
11. Is the outlier explainable by a single documented event?
12. Am I comparing compatible time units?

💡 **The Insight:** A "bad" month that coincides with a system upgrade is not a crime wave — it is a data artifact. Learn to tell the difference before escalating.

Key Takeaway: Before acting on any unusual pattern, verify that the data itself is trustworthy. Data quality is not a technical detail — it is a leadership responsibility.